

Technology Stack

Date	03 October 2026
Team ID	SWTID-2026-6945
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	2 Marks

The stack below was chosen for fast development, strong Python AI library support and simple local deployment.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Jinja2 templates, small vanilla JavaScript	Server-rendered pages need no build step; embedded CSS gives a responsive, lightweight UI (index, preview, export success).
2	Backend / Server-Side	Python 3, FastAPI, Uvicorn (ASGI), Pydantic	Fast async framework with automatic /docs, built-in validation, and the same language as the AI libraries.
3	Database / Data Storage	No database - file storage: static/panels (PNG), static/exports (PDF)	Each comic is generated on demand, so simple files are enough; keeps setup easy. A database is listed as future scope.
4	Cloud / Hosting / Deployment	Local Uvicorn server; Dockerfile provided; GPU via Google Colab / Kaggle if needed	Local deployment matches the project brief; Docker makes future cloud hosting straightforward.
5	Version Control & CI/CD	Git and GitHub	Required submission platform; supports collaboration and history. Automated tests can be added to GitHub Actions.
6	Third-Party APIs / Other Tools	Google Gemini (google-genai SDK): gemini-2.5-flash for the outline and gemini-3.5-flash for the story; Hugging Face Diffusers, PyTorch, Transformers: Stable Diffusion v1.5; FPDF2; Pillow; python-dotenv; pytest	Gemini Flash creates the structured outline, Gemini Pro writes narration and dialogue, Stable Diffusion draws panels, FPDF2 builds the PDF, Pillow saves images and placeholders.